# Wk7.1 Decision trees

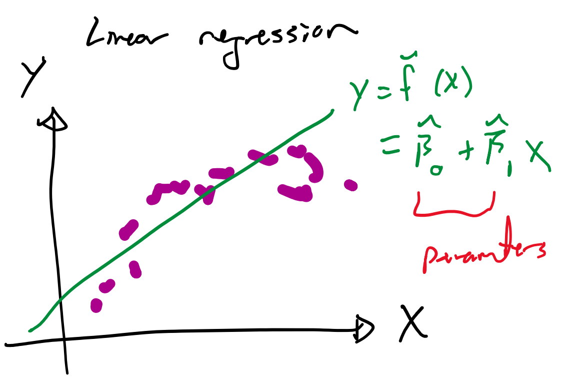
Tuesday, October 20, 2026 9:19 AM

← Ch8

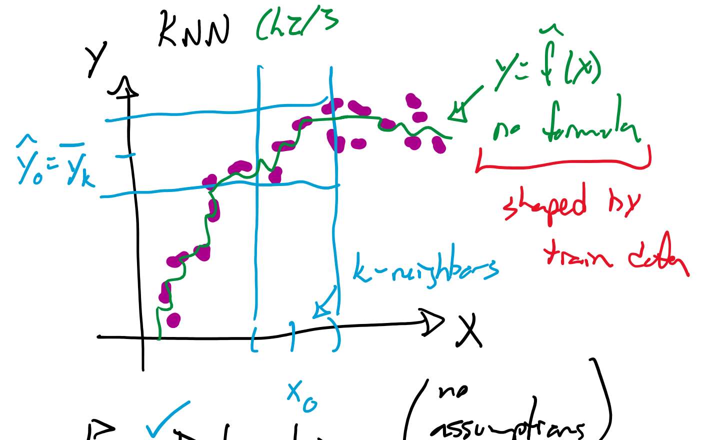
2 broad categories of methods in SL: Regression case $Y = f(X) + \epsilon$

Parametric vs Nonparametric

Linear regression



KNN Ch2/3



Pros Nice $\hat{f}$ formula which is interpretable & simple

Cons Linear assumption may be "wrong" (too simple) High bias

Pros ✓ Data-driven (no assumptions) & tunable (k via CV)

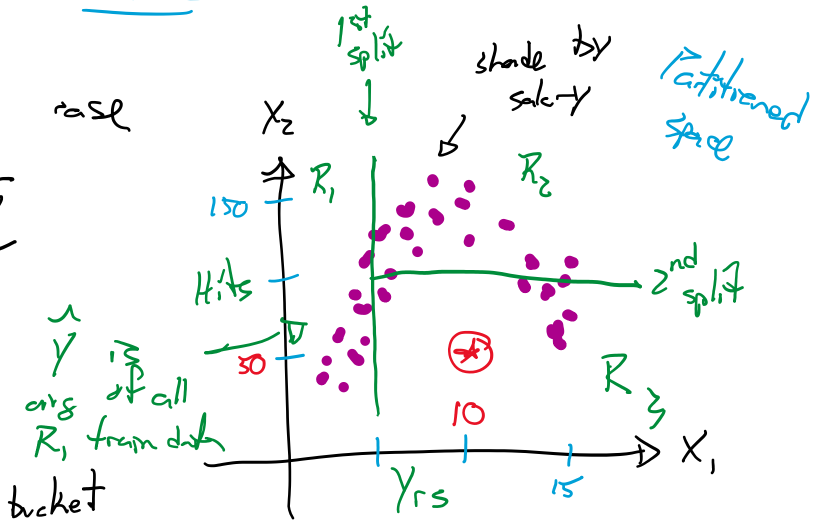
Cons Hard to interpret, sensitive to train data High variance

Add decision trees to non-parametric methods.
Improve on KNN (interpretable)

## Decision trees Regression case

Hitters salary → $Y = f(X) + \epsilon$

$X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$

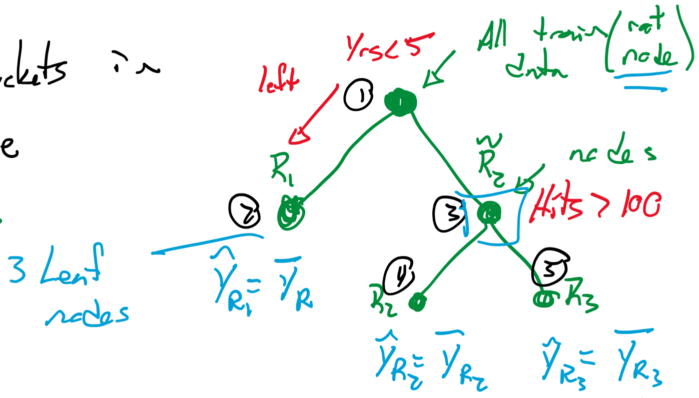


High level DT idea:

1. ① All data in one bucket
2. ② Split into 2 buckets in best way possible
   Predict & min RSS

All possible $X_1$ & $X_2$ splits

3. ③ Predict $\hat{y}_{R_j} = \bar{y}_{R_j}$
4. ④ Iterate & split more until stop condition is met
   - ⓘ Min leaf node size (other heuristic rules)
   - ⓘⓘ Split completely (each leaf 1 sample) & prune up via CV.


Yrs<5
left
All train data (root node)
①
$R_1$
②
$\hat{y}_{R_1} = \bar{y}_R$
3 Leaf nodes
$R_2$
nodes
Hits > 100
③
④
$R_2$
⑤
$R_3$
$\hat{y}_{R_2} = \bar{y}_{R_2}$
$\hat{y}_{R_3} = \bar{y}_{R_3}$

Interpretable
① Root split is most sign (Yrs matter most)
② Subtree gives subset many (old vs young)

## Detailed Version

$Y = f(X) + \epsilon, \quad X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_p \end{bmatrix}$.

All X train data is in region $R_{full}$

① Consider all possible partitions of $R_{full}$

$$R_1(j, s) = \{X \mid X_j < s\}$$

(A) predictor to cut — threshold

$$R_2(j, s) = \{X \mid X_j > s\}$$

Loop over all $X_j$'s & all threshold cuts s (all midpoints)

② Per partition possibility, compute predictions

$$\hat{Y}_{R_1} = \bar{Y}_{R_1} \quad \& \quad \hat{Y}_{R_2} = \bar{Y}_{R_2}$$

indicator fcn 0, false 1, true

$$\& \quad \hat{Y} = \hat{f}(X) = \sum_{i=1}^{2} \hat{Y}_{R_i} \underline{I(X \in R_i)} = \sum_{i=1}^{2} \bar{Y}_{R_i} I(X \in R_i)$$

③ Compute RSS per partition.

$$RSS(j, s) = \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2$$

④ Choose partition (A) which minimizes RSS

⑤ Repeat for 2 new nodes

⑥ Iterate until stop condition.

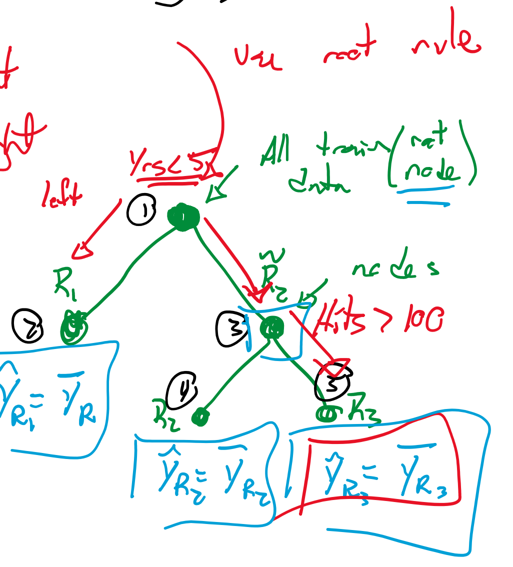
Once a model is trained, how to predict & interpret?

Predict
New player: $X_0 = \begin{bmatrix} X_{10} \\ X_{20} \end{bmatrix} = \begin{bmatrix} 10 \\ 50 \end{bmatrix}$, $\hat{Y}_0 = ?$

① yrs = 10 > 5, move right

③ Hits = 50 < 100, move right

Predict $\hat{Y}_0 = \hat{Y}_{R_3} = \bar{Y}_{R_3}$


Use root rule
Yrs<5
left
All train data (root node)
①
$R_1$
②
$\hat{y}_{R_1} = \bar{y}_R$
$R_2$
nodes
Hits > 100
③
④
$R_2$
⑤
$R_3$
$\hat{y}_{R_2} = \bar{y}_{R_2}$
$\hat{y}_{R_3} = \bar{y}_{R_3}$

Depth ↓

Interpret
Top split most important, less as go down. Quantify by RSS gains.

## Balance overfitting & underfitting

① CV ✓

(A) ② Regularization

Ch6: $\hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \dots + \hat{\beta}_p X_p$

Ridge: RSS + Shrinkage = $\sum (Y_i - \hat{Y}_i)^2 \boxed{+ \lambda \sum_{i=1}^{p} \beta_i^2}$

(A) LASSO: " = " $\boxed{+ \lambda \sum_{i=1}^{p} |\beta_i|}$

penalize big $\beta_j$'s

→ Balance RSS & Shrinkage (of leaves)

$$\sum_{i=1}^{n} (Y_i + \hat{Y}_i)^2 + \underline{\text{shrinkage}}$$

# leaves

# of leaves in T

$$= \sum_{m=1}^{|T|} \sum_{X_i \in R_m} (Y_i - \hat{Y}_{R_m})^2 + \alpha |T|$$

prune parameter to tune via $\alpha$.

Note ⓘ
Partitioning X via DTs is not realistic for regression (& classification).
Linear regress often is more accurate.
Instead use ensembles. (1000s of trees subset train data & avg predictions) to improve accuracy & beat parametric models.

samples & $X_i$'s

Random forest (Bagging & Boosting & randomly)
(A) (GBM XGBoost CATBoost) (A)

## Decision tree for classification

Almost the same as DT for regression, but need a new way to decide splits.

Predict majority class in region.
Split to improve accuracy of predictions.
Easiest, poor class imbalance & multiclass.

(A) ① Gini coef TBC
② Entropy